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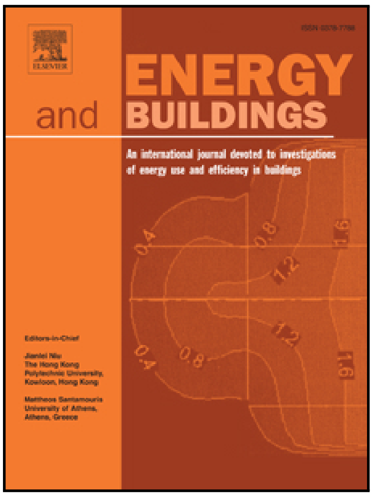

Fine-Tuned Large Language Models as Surrogate Schedulers for Smart Building Energy Management

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Highlights

• A fine-tuned LLM is proposed as a decision surrogate for MILP-based energy management in Net-Zero Energy Buildings (NZEBs); novel to this application domain. • MILP-generated optimal schedules are used to supervise LLM prediction of DG dispatch, ESS charging/discharging, grid trading, and heating/cooling device set-points. • The proposed framework is evaluated in terms of prediction accuracy, feasibility of predicted schedules, and resulting building energy system operational performance. • The results demonstrate the potential of fine-tuned LLMs to reduce the computational burden of repeated online MILP scheduling for deployment-oriented NZEB operation.

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Fine-Tuned Large Language Models as Surrogate Schedulers for Smart Building Energy Management

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A B S T R A C T

Buildings account for a substantial share of global energy consumption and associated carbon emissions, which has increased interest in net-zero energy buildings (NZEBs) and their efficient operation. In NZEB energy management, operational scheduling must coordinate distributed generation (DG), energy storage systems (ESSs), utility-grid power exchange, and thermal devices while minimizing operating cost under variable renewable generation and demand. Although mixed-integer linear programming (MILP) can provide optimal schedules, repeated online optimization may impose a considerable computational burden in real-time applications. This paper proposes a fine-tuned large language model (LLM) framework as a surrogate for MILP-based NZEB scheduling, with the aim of providing computationally efficient decision support and a flexible architecture for future smart-building energy management. The MILP model is first used to generate optimal set-points for DG dispatch, ESS charging/discharging power, grid buy/sell power, and heating/cooling device operation. These optimization-derived input-output pairs are then used to fine-tune the LLM so that it can directly predict near-optimal operational decisions without repeatedly solving the MILP problem. The proposed framework is evaluated against benchmark learning models using the same prediction-error metrics reported in the Results section. The results show that the fine-tuned LLM achieves the best overall performance, with a test MAE of 0.0853 and RMSE of 0.286, outperforming both ANN and XGBoost under the same evaluation setting. Compared with ANN, the fine-tuned LLM reduces MAE and RMSE by 88.2% and 33.3%, respectively, while also improving over XGBoost by 79.7% in MAE and 24.7% in RMSE. In addition, the predicted schedules are assessed in terms of operational feasibility and system-level performance. These findings indicate that fine-tuned LLMs can serve as computationally efficient surrogates for near-optimal NZEB control.

Nomenclature

Abbreviations

AC	Absorption chiller
ANN	Artificial neural network
BES	Building energy systems
DG	Distributed generators
EC	Electric chillers
ESS	Energy storage systems
GPT	Generative pre-trained transformer
HP	Heat pumps
LLM	Large language model
MAE	Mean absolute error
MILP	Mixed-integer linear programming
ML	Machine learning
NZEB	Net-zero energy buildings
PV	Photovoltaic systems
RES	Renewable energy sources
RMSE	Root mean square error

Indices

t, i, k	Time, DG, and PV indices

Variables

$P_{PV,k}(t)$	Output power of PV k at time t (MW).
$P_{PV}^{Total}(t)$	Total output power of the PV system at time t (MW).
$P_{DG,i}(t)$	Output power of DG i at time t (MW).

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$P_{ESS}^{Ch/Dis}(t)$	Charging/discharging power of the ESS at time t (MW).
$SOC(t)$	State of charge of the ESS at time t (MWh).
$P_G^{buy}(t)$	Buying power from the utility grid at time t (MW).
$P_G^{sell}(t)$	Selling power to the utility grid at time t (MW).
$P_L(t)$	Electric load demand at time t (MW).
$P_{HP}(t)$	Power consumption by the electric HP at time t (MW).
$P_{BL}(t)$	Power consumption by the boiler at time t (MW).
$P_{EC}(t)$	Power consumption by the EC at time t (MW).
$H_{HP/BL}(t)$	Heat output of the heat pump/boiler at time t (MWh).
$H_L(t)$	Heat load at time t (MW).
$H_{AC}(t)$	Heat consumption by the AC at time t (MWh).
$C_{AC/EC}(t)$	Cooling output of the AC/EC at time t (MWh).
$C_L(t)$	Cooling load at time t (MW).
$\pi_G^{Trade}(t)$	Trading price with the utility grid at time t (\$/MWh).

Constants

α, β	Weight coefficients of the objective function.
$P_{DG,i}^{\min/\max}$	Minimum/maximum output power of DG i (MW).
$E_{ESS}^{\max}$	Maximum energy capacity of the ESS (MWh).
$\eta_{Ch/Dis}$	Charging/discharging efficiency of the ESS.
$\eta_{e2h}^{HP/BL}$	Electric-to-heat ratio of the heat pump and boiler.
η_{e2c}^{EC}	Electric-to-cooling ratio of the electric chiller.
η_{h2c}^{AC}	Heat-to-cooling ratio of the absorption chiller.
C_i	Operating cost of DG i (\$/MWh).
ϵ_i	Maximum number of re-query attempts for sample i

1. Introduction

Over the next two decades, building energy consumption is expected to increase by more than 40%, further increasing electricity demand and the environmental impacts associated with building operation. Improving energy efficiency in buildings has therefore become a critical component of sustainable energy planning and carbon-emission reduction [1]. In this context, Building Energy Systems (BES) play a central role by coordinating energy generation, storage, conversion, and consumption to enhance operational efficiency and overall system performance [2]. Among emerging solutions, Net-Zero Energy Buildings (NZEBs) have attracted growing attention as a promising pathway toward low-carbon building operation. By balancing on-site renewable energy sources (RES) with building energy demand, NZEBs aim to reduce dependence on external energy supply while mitigating environmental impacts [3,4].

Despite their potential, the practical operation of NZEBs remains challenging due to the intermittent nature of RES, time-varying load demand, and the need to coordinate multiple energy assets under operational constraints. These challenges extend beyond energy forecasting alone and directly affect scheduling, control, and decision-making within BES operation. Although predictive models, including hybrid machine-learning approaches and photovoltaic (PV) forecasting methods, have improved the accuracy of renewable generation estimation [5], significant uncertainty still remains under variable weather and operating conditions, complicating real-time NZEB management [6]. As a result, recent studies have increasingly focused on optimization-based and uncertainty-aware energy management strategies to improve operational robustness, economic performance, and system flexibility. For example, [7] proposed a hierarchical RL–MPC framework for clusters of buildings, where learning-based MPC optimizes local HVAC set-points and district-level RL coordinates shared thermal and electrical resources, leading to improved flexibility, energy cost, self-consumption, and thermal comfort under both typical and extreme conditions. In addition, [8] developed a computationally efficient strategy for predicting dynamic temperature drop and heat loss in domestic hot water systems, enabling adaptive set-point scheduling that improves energy efficiency and reduces operating cost. Moreover, [9] introduced a temporal-online hybrid occupancy prediction framework that combines context-enriched machine learning with dynamic retraining, demonstrating improved robustness and predictive accuracy under sudden shifts in occupancy conditions.

Alongside optimization-based building operation, data-driven surrogate models have also gained increasing attention as a means to reduce computational burden and support fast decision-making in BES applications. Machine-learning techniques are particularly attractive because they can capture complex and nonlinear relationships among building states, environmental conditions, and operational responses [10,11]. Among these methods, Artificial Neural Networks (ANNs) have been widely used in building energy prediction and control to approximate nonlinear system behavior and support renewable-energy utilization [12]. Nevertheless, ANN-based models typically require extensive offline training and often need repeated retraining under changing operating conditions, which limits their adaptability in dynamic real-world settings. Moreover, their inference process is often difficult to interpret, which reduces transparency and may hinder trust and adoption by building operators and energy practitioners. Their practical use is also constrained by the time-consuming process of collecting, cleaning, and annotating data from intelligent building terminals. These limitations have encouraged continued exploration of alternative surrogate decision models that can preserve predictive performance while offering greater adaptability, interpretability, and implementation flexibility in building energy management systems.

Beyond ANN-based models, other machine-learning methods, including tree-based approaches, have also shown strong performance in structured building-energy applications because of their efficiency, robustness, and suitability for numerical tabular data. In this context, the use of Large Language Models (LLMs) in the present work is not based on the assumption that LLMs are universally superior to conventional machine-learning models for all BES tasks. Rather, the motivation for adopting a fine-tuned LLM in this study is based on several practical considerations. First, fine-tuning a pretrained model can reduce the amount of task-specific data and training effort required compared with building a model from scratch, while still providing strong predictive capability under uncertain operating conditions, including variability associated with RES. Second, it avoids the need to design highly specialized model architectures and training pipelines for each new application, instead enabling adaptation of a pretrained foundation model to the NZEB operational setting through an efficient fine-tuning process [13,14]. In addition, LLMs provide a flexible framework for representing structured operating states, exogenous inputs, and control actions in a unified and extensible form, which makes them promising candidates for surrogate decision modeling. Their relevance in BES therefore lies not only in predictive performance, but also in their adaptability, transferability, and potential to accommodate richer contextual information and broader automation workflows in smart building environments [15–17].

Recent literature has shown growing interest in applying LLMs to building-energy and related energy-system problems; however, these studies have largely focused on objectives different from the one pursued in this work. Existing contributions can be broadly grouped into three categories. The first includes LLM-based forecasting and prediction studies, where large models are used for time-series estimation, load prediction, or occupant-related behavioral modeling [18–22]. The second includes LLM-based support and recommendation frameworks, such as building-energy analytics, retrofit recommendation, and knowledge-assisted decision support [23,24]. The third includes NLP-oriented and information-management applications, such as operation and maintenance query systems, simulation support, knowledge extraction, and human-centered semantic integration in building workflows [25–28]. Although these studies demonstrate the broad potential of LLMs in building-energy applications, they primarily target forecasting, conversational support, descriptive analytics, or general automation rather than learning optimization-derived operating decisions for direct BES scheduling and control.

In contrast to prior LLM-based studies centered on forecasting, recommendation support, or generic building analytics, this paper develops a fine-tuned LLM as a surrogate decision model that learns the mapping from BES operating conditions and exogenous inputs to MILP-derived operational actions for NZEB energy management. In the proposed framework, a mixed-integer linear programming (MILP) model is first used to generate optimal operating schedules, which are then used as supervisory labels for fine-tuning the LLM. The resulting model is not intended as a generic forecasting tool, but rather as a decision surrogate that approximates optimization-based scheduling actions, including dispatch and grid-interaction decisions, in a computationally efficient manner. This formulation positions the LLM as a new surrogate option for building energy management while preserving the energy-systems focus of the problem. Accordingly, the contribution of this work lies not only in predictive accuracy, but also in demonstrating a practical scheduling layer for coordinated electrical, heating, and cooling operation in NZEBs. By learning optimization-derived operating patterns from the supervisory MILP model, the proposed framework improves the practicality and computational tractability of repeated BES scheduling. This is particularly relevant for future smart-building applications with many decision variables and frequent scheduling requirements, where direct online optimization may become computationally burdensome. The proposed framework thus supports scalable and adaptive decision-making in multi-energy smart building environments. This manuscript offers the following contributions:

- A fine-tuned LLM framework is proposed as a decision surrogate for energy management in NZEBs, where the model approximates the mapping from BES operating conditions to MILP-derived control actions rather than performing conventional forecasting, representing a novel application of LLMs in this domain.
- A MILP-based optimization layer is employed to generate optimal schedules for major BES decision variables, including DG dispatch, ESS charging/discharging power, grid buy/sell power, and heating/cooling device set-points, which are then used as supervisory labels for LLM fine-tuning.
- A comprehensive evaluation framework is established to assess the fine-tuned LLM in terms of prediction accuracy, feasibility of predicted schedules, and resulting BES operational performance, thereby extending the assessment beyond standard regression-based metrics alone.
- The study demonstrates the potential of fine-tuned LLMs to reduce the computational burden of repeated online MILP scheduling in NZEB energy management and establishes a generalized decision-surrogate framework for BES operation that supports future automation and AI-assisted energy management, while retaining the possibility of supervisory validation or feasibility-repair mechanisms for practical deployment.

This paper is structured as follows: Section 2 describes the system architecture, while Section 3 discusses the motivation for integrating LLMs into BES operation. Section 4 presents the proposed framework, which combines an MILP-based mathematical model with a fine-tuned LLM prediction model. Section 5 provides a detailed case study, including the initial data, system characteristics, tokenization process for GPT, and the corresponding results. Section 6 discusses the performance of the proposed framework, its limitations, and its deployment implications. Finally, Section 7 concludes the paper and outlines directions for future research.

2. System Architecture

The proposed framework is developed for a medium-sized commercial office building operated as an NZEB. This building type is selected as the representative case study throughout the manuscript and is used consistently in the architecture, optimization, and simulation analyses [29]. The considered BES must satisfy the building's electric, heating, and cooling demands while coordinating local distributed energy resources and grid interaction in an efficient and reliable manner. In this study, the load profiles correspond to an office-building application, and the operational setting is therefore aligned with commercial occupancy conditions and office-type daily demand patterns. The scheduling problem is formulated over a day-ahead horizon with hourly resolution, where the BES decisions are updated based on the available demand and renewable-generation information.

The operation of NZEBs remains challenging because building demand, renewable energy availability, and grid interaction must be coordinated under time-varying and uncertain conditions. In particular, uncertainty in PV generation, combined with variations in electric, heating, and cooling loads, makes real-time scheduling and energy balancing inherently complex. Consequently, the BES must simultaneously ensure energy supply adequacy, reduce operating cost, and enhance the effective utilization of local renewable resources. These requirements motivate the integrated architecture considered in this work, in which multiple energy resources and conversion devices are coordinated through a centralized energy-management framework.

The system architecture, together with the BES configuration, is illustrated in Fig. 1. The building is equipped with local PV generation, distributed generators (DGs), and an energy storage system (ESS), all of which contribute to meeting the electrical demand. The BES is also connected to the utility grid, which supplies supplementary electricity during periods of local resource insufficiency and can absorb excess power when economically beneficial. In addition to the electrical subsystem, the building includes heating and cooling devices to satisfy thermal demands. Specifically, the heating subsystem consists of a boiler and an electric heat pump (HP), while the cooling subsystem includes an electric chiller (EC) and an absorption chiller (AC).

The thermal and electrical subsystems are jointly coordinated within the BES. Electricity generated by the PV system, DGs, ESS, and utility grid is allocated not only to the building's direct electrical loads but also to electrically driven thermal devices such as the HP and EC. The boiler and HP supply the required heating energy, while the AC utilizes thermal energy to contribute to cooling production. The EC and AC therefore operate in a complementary manner to satisfy the cooling demand. Through this integrated configuration, the BES enables coordinated multi-energy operation across electrical, heating, and cooling domains.

The overall operation of the system is managed by the building energy management system, which determines the optimal scheduling of generation units, storage charging/discharging actions, grid power exchange, and thermal-device operation. This coordinated control structure improves system flexibility, reduces reliance on the external grid, and enhances the efficiency and resilience of building operation. Accordingly, the architecture provides the physical and operational foundation for the proposed MILP-based scheduling model and its fine-tuned LLM-based surrogate decision framework.

3. Motivation for Fine-Tuned LLM Integration

The use of a fine-tuned LLM in this study is not motivated by the task being inherently linguistic. Rather, the problem considered here is fundamentally a structured numerical decision-mapping task, in which the model learns the relationship between BES operating states, exogenous inputs, and optimization-derived control actions. Accordingly, language modeling is not essential in a strict sense for solving the problem. The motivation for adopting a fine-tuned LLM instead lies in the use of a

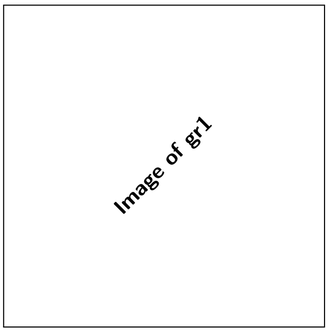

Fig. 1. Overview of the system architecture, including building energy system.

pretrained foundation model that can be adapted efficiently and that offers a flexible interface for future building energy management settings involving not only numerical variables, but also richer contextual information such as operating instructions, user preferences, demand-response signals, supervisory feedback, or other external context that may become relevant in practical building energy management applications [15–17].

At the same time, the present task remains a structured numerical problem, and lighter machine-learning models designed for tabular data, such as tree-based methods, are appropriate and relevant benchmarks. Therefore, this work does not assume that LLMs are inherently or universally superior to conventional structured-data models for BES scheduling tasks. Instead, the objective is to examine whether a fine-tuned pretrained model can provide competitive predictive performance while also offering greater flexibility for future smart-building applications involving mixed numerical and contextual inputs.

Moreover, the motivation for this framework is tied to the operational complexity of NZEBs. These systems must coordinate renewable generation, ESS operation, grid interaction, and thermal-device scheduling under uncertainty in demand and RES availability. While conventional data-driven models, including ANNs and tree-based predictors, can approximate optimal control actions effectively, but they are often developed for narrowly defined structured inputs and may offer less flexibility when broader contextual information must be incorporated. In contrast, the fine-tuned LLM is treated here as a transferable surrogate decision model built on a pretrained architecture, with the goal of approximating optimization-derived BES actions in a computationally efficient and extensible manner rather than serving as a generic natural-language model.

In the proposed framework, an MILP model is first used to generate optimal set-points for key decision variables, including DG dispatch, ESS charging/discharging power, grid exchange, and heating/cooling device operation. These optimization outputs are then used as supervisory labels for fine-tuning the LLM. Therefore, the role of the LLM is not to replace the physical or operational logic of the BES, but to learn an approximation of the decision policy encoded by the MILP solutions. This formulation enables fast inference once training is completed and supports the use of the model as a surrogate scheduler for repeated operational decision-making.

It is also important to clarify the role of operational constraints in the proposed prediction framework. The predictor does not explicitly enforce BES constraints in an analytical or hard-constrained optimization sense. Instead, feasible behavior is learned implicitly from the MILP-generated training labels, which satisfy the original operational constraints by construction. As a result, constraint satisfaction by the fine-tuned LLM is empirical rather than guaranteed. The quality of predicted schedules must therefore be assessed through post hoc feasibility analysis, including evaluation of generator operating bounds, ESS state-of-charge limits, and electric and thermal balance residuals as discussed in Section 5. This distinction is important because the model acts as a learned surrogate of optimization-derived actions, not as a constrained optimizer itself.

Under this interpretation, the fine-tuned LLM is adopted as a practical and extensible surrogate decision model for NZEB operation. Its value in the present study lies in approximating MILP-derived scheduling actions with low online computational burden, while retaining the possibility of future expansion toward richer operating environments that include mixed structured, numerical, and textual information. This motivation is more precise than treating the task as a conventional language-generation problem and better aligns the use of the model with the underlying BES energy-management objective.

4. Proposed LLM-Based Energy Management Framework

Fig. 2 illustrates the overall workflow of the proposed LLM-based energy management framework. The training phase consists of three main stages: 1) dataset collection, 2) data preprocessing, and 3) LLM training and fine-tuning.

In the first stage, an MILP-based multi-objective optimization model is used to determine the optimal setpoints for each controllable resource, aiming to minimize the BES operating costs while ensuring that all energy demands (electricity, heating, and cooling) are met. Decision variables, such as the generation outputs of the boiler, HP, chillers, DGs, and the charging/discharging levels of the ESS, are optimized. At this stage, the uncertainties in PV generation and load demand (P_L, H_L, and C_L) are taken into account. The complete mathematical formulation, including the objective function and operational constraints, is presented in Section 4.1 to ensure optimal performance. Furthermore, system-level power constraints are incorporated to enhance reliability and maintain secure BES operation. In the second stage, data preprocessing is performed to format the inputs appropriately for LLM training, as detailed in Section 4.2. In the third stage, the selected transformer model is fine-tuned using the prepared dataset so that it can learn the mapping between system operating conditions and optimization-derived control actions.

After training, the model is evaluated in the testing phase. The predicted outputs are converted back to physical units and checked against key operational limits, including DG power bounds and ESS constraints such as SOC and charge/discharge limits. If a predicted solution violates these requirements, the model receives a structured violation summary and is queried again, up to ϵ_i times for each sample. The final accepted predictions are then used to compute performance metrics such as MAE and RMSE, which are used to evaluate model performance and guide hyperparameter refinement. The complete procedure is discussed in detail in Section 4.3.

4.1. MILP-Based Mathematical Optimization Model

This section presents a detailed MILP-based mathematical model with a multi-objective function aimed at minimizing building energy

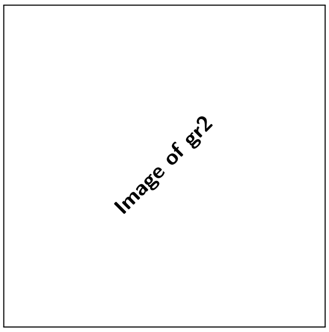

Fig. 2. General algorithm of the proposed method for fine-tuning large LLMs.

operational costs and reducing dependence on the utility grid. Additionally, the operational constraints of DGs, ESSs, and energy balances across electrical, heating, and cooling systems are also presented in detail. Eq. (1) defines the objective function, which seeks to optimize the operation of the BES. In this equation, the first term represents the operating costs of the entire BES, while the second term captures the building's reliance on the external utility grid. For each time interval, the model calculates both the cost of power generated by the DGs and the cost/profit associated with trading power with the utility grid.

$$\min \quad \sum_{t=1}^{T} \Big[\alpha \Big(C_i \cdot P_{DG_i}(t) + \pi_G^{Trade}(t) \cdot \big(P_G^{buy}(t) - P_G^{sell}(t) \big) \Big) + \beta \cdot P_G^{buy}(t) \Big] \tag{1}$$

where Eq. (2) imposes the constraint on the objective-function weighting coefficients, requiring their sum to be equal to one. The coefficients α and β are introduced to define the relative priority of the operating objectives considered in the supervisory MILP formulation. In the present study, they are kept fixed to ensure a consistent optimization reference for dataset generation and fair comparison among the predictive models. Eqs. (3)-(6) represent the operational constraints of the BES. Eq. (3) ensures that the output power generated by the DGs remains within the specified minimum and maximum limits. Eqs. (4) and (5) impose limits on the ESS charging and discharging capacities: Eq. (4) restricts the charging power to the maximum allowable rate, accounting for the current state of charge (SOC) and charging loss, while Eq. (5) limits the discharging power based on the current SOC and discharging loss. Eq. (6) updates the SOC of the ESS at each time interval t, based on the remaining energy from the previous time slot and the actual charging or discharging power. Eq. (7) calculates the total generation of the PVs within the system. Lastly, Eq. (8) enforces power balance, requiring that the total power generated from PVs, DGs, ESS discharging, and grid purchases matches the total load, which includes the building load, electric consumption of the HP, boiler, and EC, as well as any power exported to the utility grid. The heat produced by the HP and boiler must be sufficient to meet both the heat load and the heat required by the AC, as specified in Eqs. (9) and (11). Additionally, the cooling load must be fully supplied by the cooling generated from the AC and EC, as detailed in Eqs. (12) and (14).

$$\alpha + \beta = 1 \tag{2}$$

$$P_{DG\_i}^{min} \leq P_{DG\_i}(t) \leq P_{DG\_i}^{max} \qquad \forall t, \forall i \in \{1, 2, 3, \ldots, I\} \tag{3}$$

$$0 \leq P_{ESS}^{Ch}(t) \leq E_{ESS}^{max} \cdot \left(\frac{1 - SOC(t)}{\eta_{Ch}} \right) \qquad \forall t \tag{4}$$

$$0 \leq P_{ESS}^{Dis}(t) \leq E_{ESS}^{max} \cdot SOC(t) \cdot \eta_{Dis} \qquad \forall t \tag{5}$$

$$SOC(t+1) = SOC(t) + P_{ESS}^{Ch}(t) - P_{ESS}^{Dis}(t) \qquad \forall t \tag{6}$$

$$P_{PV}^{Total}(t) = \sum_{k=1}^{K} P_{PV\_k}(t) \quad \forall k \in \{1, 2, 3, \ldots, K\}, \forall t \tag{7}$$

$$\begin{aligned} & P_{PV}^{Total}(t) + P_{DG_i}(t) + P_{ESS}^{Ch}(t) - P_{ESS}^{Dis}(t) \\ & \quad + P_G^{buy}(t) - P_G^{sell}(t) \\ & = P_L(t) + P_{HP}(t) + P_{BL}(t) + P_{EC}(t) \qquad \forall t \end{aligned} \tag{8}$$

$$H_{HP}(t) + H_{BL}(t) = H_L(t) + H_{AC}(t) \qquad \forall t \tag{9}$$

$$H_{HP}(t) = \eta_{e2h}^{HP} \times P_{HP}(t) \qquad \forall t \tag{10}$$

$$H_{BL}(t) = \eta_{e2h}^{BL} \times P_{BL}(t) \qquad \forall t \tag{11}$$

$$C_{AC}(t) + C_{EC}(t) = C_L(t) \qquad \forall t \tag{12}$$

$$C_{AC}(t) = \eta_{h2c}^{AC} \times H_{AC}(t) \qquad \forall t \tag{13}$$

$$C_{EC}(t) = \eta_{e2c}^{EC} \times P_{EC}(t) \qquad \forall t \tag{14}$$

4.2. Data Preprocessing, Conversion, and Encoding for LLM Fine-Tuning

Following the solution of the MILP-based optimization model described in the previous section, the training dataset is prepared as illustrated in Fig. 2. This subsection outlines the key steps in the initial stage of fine-tuning, which include data preprocessing, encoding, and prompt engineering. The prediction task considered in this study is fundamentally a structured decision-mapping problem rather than a natural-language generation task. Specifically, the model learns the relationship between BES operating conditions and the corresponding MILP-derived optimal set-points. Nevertheless, an LLM fine-tuning approach is adopted because it provides a pretrained foundation model, enables efficient adaptation with relatively small task-specific datasets, supports a structured prompt-to-JSON decision interface, and offers future compatibility with heterogeneous and context-rich inputs, including user-level interaction and textual supervisory information. During preprocessing, all input and output variables are normalized using z-score standardization,

$$x_{\text{norm}} = \frac{x - \mu}{\sigma}, \tag{15a}$$

$$x = x_{\text{norm}}\sigma + \mu \tag{15b}$$

where μ and σ denote the mean and standard deviation computed from the training set only. This step places variables with different physical scales on a comparable numerical range, which improves training stability and reduces scale-related bias during fine-tuning of the model. During post-processing, the predicted normalized outputs are transformed back to their physical values using the inverse operation (15b). To avoid information leakage, only current input features are used in each sample, and the training, validation, and test sets are strictly separated in time. No future information is included in the input prompt. The target outputs correspond to the MILP-optimal operating set-points at the same time step. After normalization and standardization, the processed data

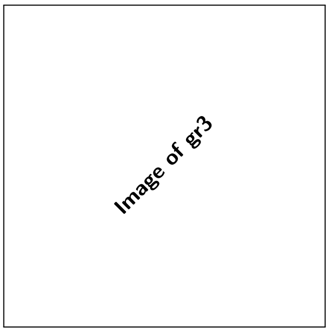

Fig. 3. Illustration of the overall procedure for the LLM fine-tuning data preparation framework.

is passed to the LLM for feature extraction and subsequent prediction of system conditions.

Unlike conventional ANN-based frameworks that often require large volumes of task-specific training data, the proposed approach fine-tunes a pretrained model using a relatively compact dataset derived from the optimization stage. The process begins by converting each numerical sample into a structured prompt format that can be processed by the LLM. In this study, a set of p-dimensional input features and q-dimensional outputs, shown in Fig. 3, is used as the general template for transforming raw data into prompt–response pairs. To clarify the prompt design, one complete example is provided below. Each training sample follows the chat-completion format and consists of three parts: a fixed system message defining the task, a user message containing the normalized BES inputs, and an assistant message containing the corresponding normalized set-points. *Example prompt–response pair:*
System message:"You are an energy management policy for a Net-Zero Energy Building (NZEB). "
"Given normalized hourly features (z-scored), produce normalized optimal setpoints!. "
"Return ONLY a compact JSON with numeric values for the required keys—no text, no units, no extra fields."
User message: "Normalized inputs:
`{"PRbuy":0.12, "PRsell":-0.34, "P_storage":0.56, "P_PV":1.2, "P_load":-0.1, "H_load":0.3, "C_load":0.8}`.
Return JSON with keys: `P_DG1, P_DG2, P_buy, P_sell, heat_boiler, cooling_AC, cooling_EC, P_dis, P_charge`."
Assistant message: `{"P_DG1":0.02, "P_DG2":0.0, "P_buy":0.15, "P_sell":-0.12, "heat_boiler":0.3, "cooling_AC":0.5, "cooling_EC":0.2, "P_dis":0.0, "P_charge":0.08}`

During inference, only the system and user messages are provided, and the model generates the assistant response. This prompt structure preserves a consistent mapping between normalized operating conditions and normalized control decisions while maintaining a compact and machine-readable output representation. Each data record is then serialized into a structured JSONL format compatible with the fine-tuning API. The input features and target variables are stored as key–value pairs to ensure schema consistency, preserve numerical precision, and facilitate seamless integration with the fine-tuning framework. The overall data-to-prompt conversion procedure is summarized in Algorithm 1.

4.3. Large Language Model Training and Fine-Tuning

After generating the prompts for fine-tuning and dividing the dataset into training, validation, and test subsets, an API key is created to access the LLM. In this study, OpenAI's GPT model is employed [30]. Fine-tuning, in the context of OpenAI's models, refers to adapting a pretrained model using a task-specific dataset to improve its performance on a particular application. During this process, the model parameters are updated with new data, allowing the LLM to capture task-specific patterns and deliver more specialized results.

Algorithm 1 Conversion of data records into JSONL prompts for LLM fine-tuning

Require: Preprocessed dataframe $\mathcal{D}$ with train/val/test splits; input column set $\mathcal{X} = [x_1, \ldots, x_p]$; output column set $\mathcal{Y} = [y_1, \ldots, y_q]$; system message SYS
Ensure: JSONL file of prompt-response pairs where each line is `{messages : [...]}`
for each record r_i in dataset $\mathcal{D}$ **do**
  Extract input features: state $\leftarrow \{x : \text{float}(r[x])\ \forall x \in \mathcal{X}\}$
  Extract target outputs: target $\leftarrow \{y : \text{float}(r[y])\ \forall y \in \mathcal{Y}\}$
  Create user prompt:
  Prompt ← "Normalized inputs: `state`. Return JSON with keys: `target`"
  Create message structure:
  Messages ← [{role : system, content : SYS}, {role : user, content : user}, {role : assistant, content : json(target)}]
  Serialize Messages as JSON object.
  Append serialized JSON to output file
end for

The fine-tuning process begins by uploading the prepared prompts into ChatGPT, where the model is trained on optimal set-points under different operating conditions. The fine-tuning job can be initiated either through the Fine-Tuning User Interface [31] or programmatically through Python using API access. Hyperparameters are then configured, and fine-tuning is executed on the selected base model. Upon completion, a unique fine-tuned model ID is generated, which can subsequently be used for inference in practical applications. Notably, the dataset size required for fine-tuning is significantly smaller than that needed to train a traditional ML model from scratch.

After fine-tuning is completed and the retrained model is obtained, the testing phase is performed with an embedded feasibility-check layer. For each test sample, the LLM first generates a predicted output vector in normalized form, which is then converted to physical units for validation. The predicted schedule is checked against operational constraints, including DG output limits and ESS-related constraints such as SOC bounds and charge/discharge limits. If the prediction satisfies these requirements, it is accepted directly. Otherwise, a structured feasibility-feedback block summarizing the detected violations is added to the original prompt, and the same LLM is queried again so that it can revise its output accordingly. However, this layer does not guarantee feasibility; rather, it attempts to encourage the LLM to revise its decision in the case of violations and keep the set-points within the system constraints. This retry process continues up to a predefined maximum number of attempts. The final accepted prediction is then used for performance evaluation. *Example of feasibility-feedback prompt in the testing phase:*
User message: "same as the original sample in Section 4.2
— Feasibility correction required (after denormalization) —
Violation: `P_DG1` breaks the DG1 power limit after denormalization (about 8.3% of steps; maximum excess approximately 12.3 kW)."

For evaluation, test prompts are constructed in the same format as the training prompts. For each test case, the fine-tuned LLM generates predicted set-points, which are recorded and compared against reference values. The predictive performance of the model is assessed using the mean absolute error (MAE) and root mean square error (RMSE), defined as follows:

$$\text{MAE} = \frac{1}{N}\sum_{i=1}^{N}\left|y_i - \hat{y}_i\right| \tag{16}$$

$$\text{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(y_i - \hat{y}_i\right)^2} \tag{17}$$

Table 1
Parameters for test system configuration in BES optimization.

Parameter	Value	Unit	Parameter	Value
$P^{min/max}_{DG\_i}$	[0,0.9]	MW	η^{HP}_{e2h}	2.0
PV Total Capacity	3	MW	$\eta_{Ch/Dis}$	0.98
E^{max}_{ESS}	1	MWh	η^{EC}_{e2c}	2.5
C_1	0.2	\$/KWh	η^{BL}_{e2h}	3.0
C_2	0.3	\$/KWh	η^{AC}_{h2c}	2.5

Table 2
Hyperparameter configuration for LLM fine-tuning.

Parameter	Value
Base model	GPT-4o-mini
Training method	Supervised
Learning rate multiplier	0.8
Batch size	3
Number of epochs	10

Table 3
Tokens and fine-tuning process for GPT-4o-mini.

Scenario	Training sample size	Trained tokens
Day	24	60,110
Week	168	420,810
Month	720	1,805,360
Year	5760	14,479,910

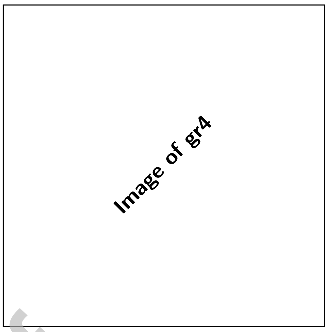

Fig. 4. Convergence of GPT-4o-mini during fine-tuning for different training set sizes: (a) training performance, (b) validation performance and (c) Minimum Validation loss.

5. Results

5.1. Case Study

This section presents a detailed analysis of the proposed prediction model and its application to the optimal operation of the BES. The outputs predicted by the fine-tuned LLM are integrated into the control framework to support more effective scheduling and management of BES operations. The prediction and optimization models are applied to a medium-sized office building located in Berlin, Germany. The electric, heating, and cooling load data are obtained from an office building model reported in [29]. The PV dataset is generated for a PV system using data from [32]. The hourly calculations are based on a realistic residential load curve extracted from a typical office building. The ANN coding, fine-tuned LLM execution through API access, and model evaluation were performed on an Intel Core i7-4790 CPU at 3.60 GHz with 16 GB of RAM, using Python in the Spyder environment. Initial test system data is presented in Table 1.

The LLM model employed in this study was fine-tuned from the GPT-4o-mini base model [30]. Table 2 summarizes the hyperparameters used for LLM fine-tuning. The hyperparameters were selected based on OpenAI API guidelines and validation-loss monitoring rather than through a systematic tuning process [31]. The chosen values resulted in stable validation convergence, and no hyperparameter ablation study was performed. The dataset was transformed into structured text prompts following the template shown in Fig. 3, ensuring compatibility with the model's tokenization process. GPT-4o-mini was intentionally selected as a baseline model to evaluate LLM performance and demonstrate the capability and generalizability of the proposed fine-tuning framework. Although larger-scale models are expected to deliver higher accuracy, GPT-4o-mini offers a cost-effective and efficient alternative, making it well-suited for validating the proposed methodology. A comparative analysis of different LLM models is provided in Section 5.4.

5.2. Impact of Training Data Size

To evaluate the influence of training data size on model performance, multiple datasets with varying sample sizes were used for fine-tuning. In general, 70% of the available data was allocated for training, while 15% of the total dataset (equivalent to 20% of the training size) was used for validation. To assess the model's ability to predict optimal setpoints for the BES, testing was performed on a 24-hour horizon for all training-size scenarios. The maximum number of re-query attempts per sample, ϵ_i, is set to 3 (including the initial query).

For fine-tuning, the model processes inputs using OpenAI's tokenizer, which converts text into tokens—subword units rather than complete words. Frequently occurring words are usually mapped to a single token, while less common or compound expressions may be split into multiple tokens. This design allows the model to efficiently represent diverse vocabularies, technical terminology, and numerical values. Tokenization is therefore essential for encoding structured inputs such as load values, setpoints, and time-series data into prompt formats. Since the computational cost of both training and inference depends on the number of tokens rather than characters, careful prompt engineering was employed in this study to reduce token count while preserving critical contextual information. Table 3 presents the fine-tuning details, including token usage and dataset size. For transparency, the training budget has been reported in tokens; the monetary cost depends on current pricing and can be estimated by Eq. 18. For example, with 60,110 trained tokens, the cost is approximately $0.06011 \times$ rate (USD per 1M tokens) [30].

$$\text{cost} \approx \left(\frac{\text{trained tokens}}{10^6}\right) \times \text{rate}. \tag{18}$$

Fig. 4(a) shows the training process across different data-size scenarios. As expected, larger training sets lead to faster convergence (i.e., fewer epochs to reach a given performance). Note that the number of steps in fine-tuning differs from the number of epochs. For comparability, results are reported per epoch. For example, with one month of training data (720 samples), batch size 3, and 10 epochs, the total steps are $(720/3) \times 10 = 2400$. Because step counts vary with dataset size and batch size, normalizing by epochs enables a consistent comparison across scenarios. Fig. 4(b) presents the validation loss for each scenario, and 4(c) indicating the best performance for the year-sized sample, followed by the one-month dataset. Fig. 5(a) and 5(b) report the test-set MAE and RMSE, respectively, after fine-tuning across all data-size scenarios. The one-month training set delivers markedly more reliable performance than the day and week sets, while the year-long set attains the lowest errors overall. However, the monthly configuration requires far fewer tokens (1.8 M) than the yearly configuration (14.5 M), making the one-month dataset a more cost-effective and scalable choice for subsequent analyses and validation.

5.3. Baseline Comparison

5.3.1. System Performance

In this section, ANN and XGBoost are used as benchmark prediction models to validate the outputs of the proposed fine-tuning approach for

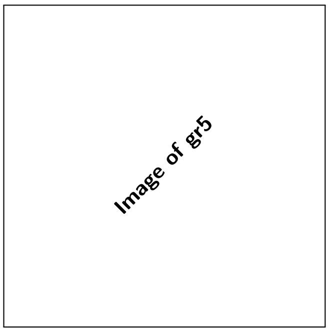

Fig. 5. Test performance across data-size scenarios using GPT-4o-mini: (a) MAE; (b) RMSE.

Table 4
Hyperparameter configurations for ANN and XGBoost models.

ANN		XGBoost	
Parameter	Value	Parameter	Value
Model	Feed-forward	Number of trees	200
Training method	Supervised	Max tree depth	6
Learning rate	1×10^{-3}	Learning rate	0.1
Activation function	ReLU	Subsample ratio	1
Optimizer	Adam	Column subsample ratio	1
Batch size	3	Minimum child weight	1
Number of epochs	10		
Hidden layers	2		
Neurons per layer	64		

optimal set-point forecasting. The hyperparameters used for ANN and XGBoost training are summarized in Table 4. XGBoost is an ensemble learning method based on gradient-boosted regression trees, in which the final prediction is generated through a sequence of decision trees [33]. At each boosting iteration, a new tree is added to minimize a regularized objective that includes both prediction error and tree-complexity penalty terms. Unlike ANN-based models, XGBoost is particularly effective for structured tabular data and can capture nonlinear interactions with relatively low training complexity. However, similar to ANN predictors, XGBoost does not explicitly enforce system constraints unless an external feasibility-repair or post-processing mechanism is applied.

The results from the fine-tuned LLM are compared with those from the ANN and XGBoost methods in Fig. 6. The fine-tuned LLM demonstrates a closer alignment with actual operating profiles across most output variables, while XGBoost also achieves strong performance and consistently improves over the ANN baseline. In the AC cooling Fig. 6(a) and heat boiler outputs Fig. 6(b), both the LLM and XGBoost are able to track the major variations and peak values more accurately than the ANN. This high performance can be attributed to the LLM's architecture, which effectively captures complex and non-linear patterns within the data. Additionally, the fine-tuned LLM shows more reliable predictions for the power exchanged between the building and the utility grid in Fig. 6(d), the generator set-points in Fig. 6(c), and the ESS charging/discharging power in Fig. 6(e). This indicates its robustness in handling energy system operations, as the LLM follows the actual trend more closely and captures operating transitions better than the ANN and slightly better than XGBoost.

Fig. 7 compares the test errors of the three prediction models and shows that the fine-tuned LLM achieves the best overall performance, followed by XGBoost, while the ANN yields the largest errors. As shown in Fig. 7(a)–(d), the LLM attains the lowest MAE, RMSE, and NRMSE values, equal to 0.0282, 0.268, and 0.383, respectively, compared with 0.139, 0.356, and 0.509 for XGBoost, and 0.238, 0.402, and 0.573 for the ANN. Relative to the ANN, the fine-tuned LLM reduces the MAE, RMSE, and NRMSE by approximately 88.2%, 33.3%, and 33.2%, respectively. Compared with XGBoost, the LLM further reduces these errors by

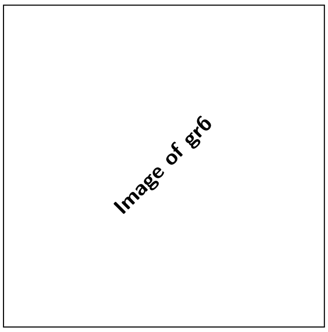

Fig. 6. Comparison of actual values and predictions from the fine-tuned LLM, ANN and XGBoost models.

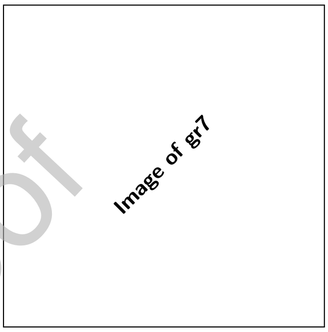

Fig. 7. Comparison of evaluation metrics across three prediction models on test data.

about 79.7%, 24.7%, and 24.8%, respectively. These results confirm that the fine-tuned LLM provides the most accurate and stable prediction of BES optimal set-points among the examined methods, while XGBoost also offers a clear improvement over the ANN baseline.

The error-distribution plots in Fig. 7(e) provide additional insight into model behavior. The LLM exhibits a more concentrated error distribution around zero, indicating more consistent predictions and fewer large deviations. XGBoost shows a similar but more dispersed pattern, whereas the ANN presents the widest spread and more pronounced extreme errors. This behavior suggests that the fine-tuned LLM is better able to capture the nonlinear relationships between system states and optimal operating decisions, leading not only to lower average error but also to more reliable prediction performance across different operating conditions. Notably, this level of accuracy is achieved using only one month of training data, which highlights the effectiveness of the proposed fine-tuning strategy under limited data availability.

5.3.2. Feasibility Assessment

Fig. 8 presents the feasibility and constraint-satisfaction performance of the actual schedule and the three prediction-based methods. It should be noted that ANN, XGBoost, and the fine-tuned LLM all act as predictive surrogates rather than constrained optimizers; therefore, none of them provides a formal guarantee of feasibility by itself. However, in the proposed fine-tuned LLM-based surrogate scheduler, a feasibility-check layer is incorporated to reduce violations by allowing the LLM to revise its decisions when constraint violations are detected. The differences observed in Fig. 8 should thus be interpreted as an empirical reflection of how effectively each model learns feasible operating patterns from the MILP-generated labels, rather than as evidence of explicit built-in constraint enforcement.

As shown in Fig. 8, the ANN exhibits the highest violation rates across several constraints, including $P_{\mathrm{DG},2}(t)$, $P_{\mathrm{ESS}}^{\mathrm{Ch/Dis}}(t)$, $P_G^{\mathrm{buy}}(t)$, $P_G^{\mathrm{sell}}(t)$, and $C_{\mathrm{EC}}(t)$. XGBoost improves the feasibility performance relative to the ANN and reduces the violation rates in most cases, but violations are still observed, particularly for the ESS charging/discharging power and grid

Table 5
Fine-tuning phase durations for different models.

Phase	gpt-4o-mini	gpt-4.1-mini	gpt-4o	gpt-4.1
Validation	0:02:57	0:02:20	0:02:21	0:02:19
Queue	0:00:03	0:00:11	13:28:47	13:50:01
Training	0:42:28	0:39:05	2:38:10	2:37:47
Policy evaluation	0:14:29	0:15:21	0:13:21	0:22:29
Wrap-up	0:00:08	0:00:08	0:00:16	0:00:17
Total (excluding queue)	0:59:54	0:56:54	2:53:52	3:02:35

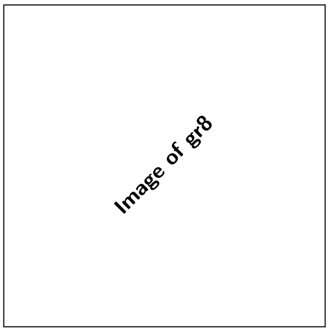

Fig. 8. Comparison of feasibility and constraint-satisfaction performance for different models.

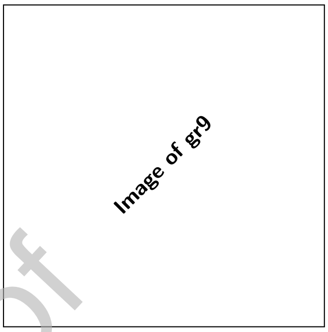

Fig. 9. Comparison of evaluation metrics (MAE and RMSE) across different LLM models.

power exchange variables. A similar trend is observed in the generator dispatch results shown in Fig. 6(c), where the ANN occasionally predicts outputs beyond the operational limits of the dispatchable units, while the LLM and XGBoost remain much closer to the feasible operating region. Among the three predictive models, the fine-tuned LLM achieves the best overall feasibility performance, with zero violations for most variables and only a limited violation rate for $P_{\mathrm{ESS}}^{\mathrm{Ch/Dis}}(t)$. Overall, the fine-tuned LLM shows the most consistent behavior across both thermal and electrical outputs, further supporting its reliability for BES operational set-point forecasting under the same training setting.

It is important to note that no hard-coded feasibility enforcement is applied to the LLM outputs. Instead, the fine-tuned LLM learns the input–output relationships directly from prompt–target training pairs. Feasibility is improved through the embedded feasibility-check mechanism, which allows the model to identify violations, reason about them, and revise its outputs accordingly. As a result, the generated schedules tend to remain closer to the feasible operating region. This behavior suggests stronger structured pattern learning from the MILP-generated data, but it should not be interpreted as proof that the LLM explicitly knows or enforces the system constraints. Similarly, the ANN may fail to satisfy some constraints under limited-data conditions, and its feasibility performance could potentially improve with more training samples or additional training epochs. However, the objective of this study is to compare all methods under the same limited-data setting and to evaluate how effectively each model can learn the complex operational relationships of the BES from the available data.

5.4. LLM Model Comparison

To assess sensitivity to the choice of base model, we fine-tuned several OpenAI models [30] using the same one-month dataset. The fine-tuning configuration (epochs, batch size, learning-rate multiplier) was held constant across models, and the total number of training tokens was identical in all cases (1,805,360; see Table 3). Table 5 summarizes practical aspects of each run, including model-specific cost, job creation metadata, and wall-clock durations for each stage of the process. Reporting is disaggregated to separate operational overhead from model-dependent training time; we also provide *Total (excluding* queue) as an effective job-time metric that is independent of transient cluster load. For clarity, we define the timing stages reported in Table 5:

- **Validation:** From job creation to "files validated." Includes data integrity, formatting, and tokenization checks; largely model-agnostic.
- **Queue:** From "files validated" to "fine-tuning job started." Reflects waiting for compute resources (cluster congestion), not model complexity.
- **Training:** From "fine-tuning job started" to "new fine-tuned model created." Supervised fine-tuning proper (checkpoints occur within this window); driven by token count, epochs, batch size, learning rate, and base-model size.
- **Policy evaluation:** From "new fine-tuned model created to "usage policy evaluations completed." Post-training safety/policy checks; typically shorter than training.
- **Wrap-up:** From "policy evaluations completed" to "job successfully completed." Final registration and enablement; seconds-level overhead.
- **Total (excluding** queue): The effective job time is defined as the sum of Validation, Training, and Policy Evaluation, and is independent of queue delays.

Fig. 9 reports the test MAE and RMSE across different base LLMs. Under an identical fine-tuning setup and a one-month training dataset, *GPT-4.1* attains the best overall predictive performance, achieving the lowest test MAE (0.003) and RMSE (0.026), followed by *GPT-4.1 mini* with a MAE of 0.007 and RMSE of 0.047. By comparison, *GPT-4o mini* and *GPT-4o* exhibit notably larger errors under the same conditions. These findings suggest that, for BES set-point prediction with limited fine-tuning data, the *GPT-4.1* family is better suited to learning the optimization-derived control patterns than the *GPT-4o* family. Aggregate results over the test horizon also favor *GPT-4.1*. The performance of these models under higher uncertainty levels is discussed further in Section 5.5.

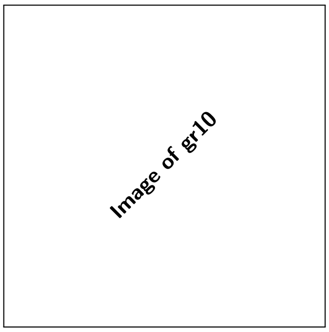

Fig. 10. Validation of different methods under input uncertainty.

5.5. Uncertainty Analysis

To evaluate robustness under input uncertainty, selected input features are perturbed using multiplicative Gaussian noise while keeping the ground-truth targets unchanged. This design isolates the effect of uncertainty in the model inputs and, for the LLM-based methods, in the numerical values embedded in the prompts. The perturbed features include grid buy price (PR_{buy}), grid sell price (PR_{sell}), PV power (P_{PV}), electrical load (P_{load}), heating load (H_{load}), and cooling load (C_{load}), while the storage-state input is not perturbed. For each feature, the perturbed value is generated as

$$x_{\text{perturbed}} = x(1+\epsilon), \tag{19}$$

where $\epsilon \sim \mathcal{N}(0, \sigma^2)$ and $\sigma \in \{0.1, 0.2, 0.3, 0.4, 0.5\}$ represents the uncertainty level. Noise is assumed independent across features and time (no temporal correlation) and homoscedastic within each level (constant coefficient of variation). For each uncertainty level, 10 Monte Carlo replications are performed, and the same noise realization is applied to all compared methods for a fair evaluation.

Fig. 10 reports the mean MAE and RMSE values, with error bars indicating the standard deviation across Monte Carlo runs. As expected, the prediction error increases as the uncertainty level rises for all methods. However, the fine-tuned LLM-based models remain more robust than the ANN and XGBoost baselines, particularly at lower and moderate uncertainty levels. Among the tested models, GPT-4o-mini and GPT-4.1-mini achieve the lowest MAE and RMSE values at 10% and 20% uncertainty, as highlighted in Fig. 10(c) and Fig. 10(d). At higher uncertainty levels, the error increases for all methods, but the newer LLM variants still maintain competitive and generally lower error levels than the conventional baselines. Overall, these results indicate that the fine-tuned LLM framework is more robust to uncertainty in BES operating inputs and can preserve prediction quality more effectively under perturbed conditions.

6. Discussion

6.1. Key Findings

This study presents a fine-tuned LLM framework for forecasting BES optimal set-points in a NZEB using supervisory labels generated by a deterministic MILP model. Rather than repeatedly solving the optimization problem online, the proposed framework learns the mapping between system operating conditions and the corresponding optimization-derived control actions. Under the same training and testing conditions, the fine-tuned LLM achieved the best overall predictive performance among the compared surrogate models, including ANN and XGBoost. The results show that the proposed LLM-based framework reduces prediction error, improves feasibility behavior, and maintains stronger robustness under input uncertainty, indicating that it can reproduce MILP-derived operating patterns more effectively than the benchmark methods.

From a building energy management perspective, the primary contribution lies not only in predictive accuracy, but also in enhancing the practical usability of optimization-based scheduling. In NZEB applications, repeated MILP-based scheduling can be computationally intensive, particularly for large-scale systems with multiple coupled variables. The proposed surrogate framework enables fast approximation of optimization-derived decisions, supporting day-ahead and receding-horizon control while preserving coordinated multi-energy operation. Therefore, the approach should be viewed as an optimization-guided decision support tool rather than a direct replacement for MILP.

A notable observation is that the fine-tuned LLM performs well even under limited-data conditions. This may be attributed to the structured prompt–response formulation and the ability of the pretrained transformer model to adapt effectively to relatively small task-specific datasets. In contrast, the ANN baseline appears more sensitive to limited training samples and local over- or under-estimation, particularly for variables strongly coupled through electrical and thermal operating constraints. XGBoost improves the benchmark comparison and performs better than ANN in most cases, especially for structured tabular inputs; however, the fine-tuned LLM still provides the most consistent performance across the coupled BES outputs. These findings support the use of fine-tuned LLMs as practical surrogate decision models for optimization-guided energy management tasks.

The uncertainty analysis further supports this conclusion. When multiple input features are perturbed using multiplicative Gaussian noise with temporally correlated errors, prediction error increases for all methods, as expected. Nevertheless, the fine-tuned LLM remains more robust than the conventional baselines across most uncertainty levels. In addition, the feasibility analysis shows that the LLM outputs remain closer to the feasible operating region than ANN and XGBoost in the reported cases. This improvement should be interpreted as an empirical result of more effectively learning feasible operating patterns from MILP-generated supervisory labels, rather than as evidence of explicit constraint enforcement. Although the reduction in aggregate error metrics such as MAE and RMSE may appear moderate in isolation, its operational significance is meaningful, since errors in ESS power, grid exchange, and thermal set-points can propagate into cost deviations and energy imbalances. Therefore, the improved accuracy of the LLM translates into better alignment with optimal operational strategies and enhances its suitability for real-world energy management applications.

6.2. Limitations

Despite the encouraging results, several limitations should be acknowledged. The proposed framework does not provide a formal feasibility guarantee, since the LLM, ANN, and XGBoost models act as predictive surrogates rather than constrained optimizers. In addition, the study is based on a single-building case study, and broader validation across different building types, climates, tariff structures, and BES configurations is needed to assess general applicability.

The proposed method should also be viewed in its proper methodological context. The LLM is used here as a surrogate of MILP-optimal actions rather than a direct optimizer, which enables fast inference while preserving an optimization-based supervisory reference. This is a practical choice for real-time BES scheduling, although deployment through an API-based workflow may introduce external dependence, latency, and cloud-execution considerations. In addition, the weighting coefficients in the MILP objective are fixed in this study to define a consistent supervisory policy, and a comprehensive sensitivity analysis of these coefficients is left for future work. Comparisons with open-source or non-OpenAI LLM families are also left for future investigation.

It should also be noted that the present study is based on a single medium-sized commercial office building. Therefore, the reported findings should be interpreted as case-study-based rather than universally generalizable across all building types, climates, tariff structures, and BES configurations. Nevertheless, the selected case captures the core multi-energy scheduling problem of NZEB operation, including coupled electrical, heating, and cooling demands, PV generation, ESS operation, grid exchange, and multiple controllable devices, and thus serves as a meaningful benchmark for proof-of-concept validation of the proposed framework.

6.3. Future Directions

Several directions can extend this work. One important direction is to investigate hybrid frameworks in which surrogate predictions are coupled with explicit feasibility-repair or constraint-screening mechanisms to improve operational reliability. Another promising direction is to examine whether optimization structure or physical constraints can be incorporated more directly into the fine-tuning or inference process. Although direct LLM-based optimization is outside the scope of this study, it remains an important long-term research topic.

Broader benchmarking is also needed. Future studies should evaluate additional open-source and commercial LLM families, explore systematic hyperparameter sensitivity, and examine the effect of larger and more diverse domain-specific training datasets. Further validation on multiple buildings and operating settings would help determine the generalizability of the proposed approach. Beyond NZEB scheduling, similar LLM-based surrogate frameworks may also be extended to related energy applications such as economic dispatch, demand response, and integrated ESS management, where fast approximation of optimization-derived decisions is valuable. Future work should examine the transferability of the proposed method across a wider range of smart-building settings, including different occupancy profiles, climate conditions, pricing structures, ESS capacities, and HVAC/BES configurations.

7. Conclusions

This work proposed and evaluated a fine-tuned LLM-based framework for BES set-point prediction in a NZEB environment. The results demonstrate that the proposed approach provides more accurate and robust predictions than conventional surrogate models, including ANN and XGBoost, while maintaining strong performance under input uncertainty. These findings support the use of LLM-based surrogate models as a promising tool for AI-assisted energy management, particularly for applications requiring fast and reliable decision support under limited-data conditions.

Despite these advantages, several limitations remain. The current framework does not explicitly guarantee feasibility, relies on API- and cloud-based deployment, and may be affected by latency and limited interpretability. In addition, the present study is based on a single medium-sized commercial office building. Nevertheless, the selected case captures key characteristics of NZEB operation, including the coupling between electrical and thermal subsystems, PV generation, ESS scheduling, and grid interaction, and therefore provides a meaningful benchmark for proof-of-concept validation. Future work should extend the proposed framework to other building types and energy system configurations to assess its broader applicability.

CRediT authorship contribution statement

Rouzbeh Haghighi: Writing – original draft, Investigation, Methodology, Conceptualization; **Van-Hai Bui:** Writing – review & editing, Supervision, Methodology, Validation; **Wencong Su:** Writing – review & editing, Supervision, Validation, Project administration, Funding acquisition.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary material

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References

[1] D. Mariano-Hernández, L. Hernández-Callejo, A. Zorita-Lamadrid, O. Duque-Pérez, F.S. García, A review of strategies for building energy management system: Model predictive control, demand side management, optimization, and fault detect & diagnosis, Journal of Building Engineering 33 (2021) 101692.
[2] D. Bonilla, M.G. Samaniego, R. Ramos, H. Campbell, Practical and low-cost monitoring tool for building energy management systems using virtual instrumentation, Sustainable cities and society 39 (2018) 155–162.
[3] W. Wu, H.M. Skye, Residential net-zero energy buildings: Review and perspective, Renewable and Sustainable Energy Reviews 142 (2021) 110859.
[4] P. Afzali, M. Rashidinejad, A. Abdollahi, M.R. Salehizadeh, H. Farahmand, A stochastic multi-objective model for energy efficiency and renewable resource planning in energy communities: A sustainably cost-effective trade-off, IET Renewable Power Generation 17 (5) (2023) 1078–1091.
[5] R. Kabilan, V. Chandran, J. Yogapriya, A. Karthick, P.P. Gandhi, V. Mohanavel, R. Rahim, S. Manoharan, Short-Term Power Prediction of Building Integrated Photovoltaic (BIPV) System Based on Machine Learning Algorithms, International Journal of Photoenergy 2021 (1) (2021) 5582418.
[6] Y. Pan, M. Zhu, Y. Lv, Y. Yang, Y. Liang, R. Yin, Y. Yang, X. Jia, X. Wang, F. Zeng, et al., Building energy simulation and its application for building performance optimization: A review of methods, tools, and case studies, Advances in Applied Energy 10 (2023) 100135.
[7] P. Hajialigol, P. Papadopoulos, A. Moazami, M. Aghaei, A hierarchical energy management system for a cluster of buildings: Reinforcement learning and model predictive control (RL-MPC) approach, Energy and Buildings (2025) 116879.
[8] P. Zheng, J. Liu, P. Liu, Y. Nakanishi, An optimal operational scheduling model for energy-efficient building with dynamic heat loss prediction, Energy and Buildings 280 (2023) 112735.
[9] R. Zhang, O.V. Klymenko, M. Short, Temporal-online hybrid machine learning for occupancy forecasting: Enhancing energy efficiency in smart building management systems, Energy and Buildings (2025) 116736.
[10] C. Fan, F. Xiao, C. Yan, C. Liu, Z. Li, J. Wang, A novel methodology to explain and evaluate data-driven building energy performance models based on interpretable machine learning, Applied Energy 235 (2019) 1551–1560.
[11] R. Haghighi, A. Hassan, V.-H. Bui, A. Hussain, W. Su, Deep Reinforcement Learning-Based Optimization of Second-Life Battery Utilization in Electric Vehicles Charging Stations, arXiv preprint arXiv:2502.03412 (2025).
[12] T.F. Megahed, S.M. Abdelkader, A. Zakaria, Energy Management in Zero-Energy Building Using Neural Network Predictive Control, IEEE Internet of Things Journal 6 (3) (2019) 5336–5344.
[13] Y. Hu, H. Kim, K. Ye, N. Lu, Applying fine-tuned LLMs for reducing data needs in load profile analysis, Applied Energy 377 (2025) 124666.
[14] X. Lin, W. Wang, Y. Li, S. Yang, F. Feng, Y. Wei, T.-S. Chua, Data-efficient Fine-tuning for LLM-based Recommendation, in: Proceedings of the 47th international ACM SIGIR conference on research and development in information retrieval, 2024, pp. 365–374.
[15] L. Zhang, Z. Chen, Opportunities of applying Large Language Models in building energy sector, Renewable and Sustainable Energy Reviews 214 (2025) 115558.
[16] S. Zheng, K. Pan, J. Liu, Y. Chen, Empirical study on fine-tuning pre-trained large language models for fault diagnosis of complex systems, Reliability Engineering & System Safety 252 (2024) 110382.
[17] S. Mohammadi, A. Hassan, R. Haghighi, V.-H. Bui, W. Su, Large Language Models for Solving Economic Dispatch Problem, arXiv preprint arXiv:2505.21931 (2025).
[18] Y. Zhang, D. Wang, G. Wang, P. Xu, Y. Zhu, Data-driven building load prediction and large language models: Comprehensive overview, Energy and Buildings 326 (2025) 115001.
[19] Z. Lai, T. Wu, X. Fei, Q. Ling, BERT4ST:: Fine-tuning pre-trained large language model for wind power forecasting, Energy Conversion and Management 307 (2024) 118331.
[20] S. Jadhav, B. Sevak, S. Das, A. Hussain, W. Su, V.-H. Bui, Scalable Fairness Shaping with LLM-Guided Multi-Agent Reinforcement Learning for Peer-to-Peer Electricity Markets, arXiv preprint arXiv:2508.18610 (2025).

[21] T. Zhou, P. Niu, L. Sun, R. Jin, et al., One fits all: Power general time series analysis by pretrained lm, Advances in neural information processing systems 36 (2023) 43322–43355.
[22] D. Liu, X. Zhou, Y. Li, Human-in-the-Loop: Integrating Virtual occupant behavior simulation and semantic Decision-Making for building energy efficiency, Energy and Buildings (2026) 117206.
[23] I. Papaioannou, C. Korkas, E. Kosmatopoulos, Smart Building Recommendations with LLMs: A Semantic Comparison Approach, Buildings 15 (13) (2025) 2303.
[24] T. Xiao, P. Xu, Exploring automated energy optimization with unstructured building data: A multi-agent based framework leveraging large language models, Energy and Buildings 322 (2024) 114691.
[25] Y. Li, M. Ji, J. Chen, X. Wei, X. Gu, J. Tang, A large language model-based building operation and maintenance information query, Energy and Buildings 334 (2025) 115515.
[26] L. Zhang, Z. Chen, V. Ford, Advancing building energy modeling with large language models: Exploration and case studies, Energy and Buildings 323 (2024) 114788.
[27] M. Arslan, S. Munawar, Large language models in building energy applications: a survey, Energy and Buildings (2025) 116800.
[28] M. Liu, L. Zhang, J. Chen, W.-A. Chen, Z. Yang, L.J. Lo, J. Wen, Z. O'Neill, Large language models for building energy applications: Opportunities and challenges, in: Building Simulation, 18, Springer, 2025, pp. 225–234.
[29] R. Khalil, G.V. Hollweg, A. Hussain, W. Su, V.-H. Bui, Assessment of Solar Energy Generation Toward Net-Zero Energy Buildings, Algorithms 17 (11) (2024) 528.
[30] OpenAI, 2025, (https://platform.openai.com).
[31] OpenAI, Fine-tune UI, 2025, (https://platform.openai.com/finetune).
[32] National Renewable Energy Laboratory (NREL), PVWatts Calculator, 2025, (https://pvwatts.nrel.gov/index.php).
[33] S.G.K. Uyar, B.K. Ozbay, B. Dal, Interpretable building energy performance prediction using XGBoost Quantile Regression, Energy and Buildings 340 (2025) 115815.